

InvestSmart Installation and Deployment Guide

InvestSmart Installation and Deployment Instructions

1. Required Software:

- Visual Studio Code
- XAMPP with Apache and MySQL
- Web browser such as Chrome or Edge
- Git
- Optional: ngrok for connecting a hosted frontend to a backend running on a local machine
- Optional: Vercel for frontend hosting

2. Project Folders:

Recommended XAMPP location:

C:\xampp\htdocs\investsmart

Main folders:

- public: frontend HTML, CSS, and JavaScript files
- backend: PHP backend classes and API endpoint
- database: MySQL schema and update scripts
- docs: project documents, ERD, PDFs, and guides

3. Pulling the Project from GitHub:

Open Command Prompt or Git Bash and run:

```
cd C:\xampp\htdocs
git clone https://github.com/Gorillamac/investsmart-backend.git investsmart
```

If the folder already exists:

```
cd C:\xampp\htdocs\investsmart
git pull origin main
```

4. Starting XAMPP:

1. Open XAMPP Control Panel.
2. Start Apache.
3. Start MySQL.
4. Confirm Apache is running on <http://127.0.0.1>.
5. Confirm phpMyAdmin opens at <http://127.0.0.1/phpmyadmin>.

5. Creating the Database:

Method 1: phpMyAdmin

1. Go to <http://127.0.0.1/phpmyadmin>.
2. Create a database named investsmart.
3. Select the investsmart database.
4. Open the SQL tab.
5. Copy and run database/schema.sql.

Method 2: Command Prompt

```
cd C:\xampp\mysql\bin
mysql -u root
CREATE DATABASE IF NOT EXISTS investsmart;
USE investsmart;
SOURCE C:\xampp\htdocs\investsmart/database/schema.sql;
```

6. Updating an Existing Database for OTP:

If the database existed before OTP was added:

```
USE investsmart;
SOURCE C:\xampp\htdocs\investsmart/database/update_otp_table.sql;
```

InvestSmart Installation and Deployment Guide

7. Backend Configuration:

Open backend/config.php and confirm:

- database host is 127.0.0.1 or localhost
- database name is investsmart
- username is root
- password is blank for default XAMPP setup

8. Running the Backend Locally:

Start Apache and MySQL in XAMPP.

Open this API endpoint in the browser:

`http://127.0.0.1/investsmart/backend/api/index.php`

If the backend returns JSON or an API message, PHP is running.

9. Running the Frontend Locally:

Open:

`http://127.0.0.1/investsmart/public/index.html`

Use the home page to sign up or sign in.

10. Default Admin Login:

Email: `admin@investsmart.local`

Password: `password`

After entering email and password, the system displays a demo OTP on the screen. Enter that OTP to complete login.

11. Hosted Frontend with Local Backend:

This option is used when the frontend is deployed online, but the backend runs from a student's laptop using XAMPP.

Steps:

1. Start Apache and MySQL in XAMPP.
2. **Confirm backend works locally:**
`http://127.0.0.1/investsmart/backend/api/index.php`
3. **Start ngrok:**
`ngrok http 80`
4. Copy the HTTPS ngrok forwarding URL.
5. Open `public/assets/js/env.js` in the hosted frontend project.
6. **Set the API base URL like this:**
`window.INVESTSMART_API_BASE = "https://YOUR-NGROK-LINK.ngrok-free.app/investsmart/backend/api/index.php";`
7. Commit and redeploy the frontend.
8. Test login from the deployed frontend.

12. Important ngrok Notes:

- The ngrok URL changes unless a reserved domain is used.
- If ngrok says the endpoint is offline, restart ngrok and update `env.js`.
- Use HTTPS ngrok URL, not HTTP.
- Keep XAMPP Apache and MySQL running while users test the hosted frontend.

13. Frontend Deployment on Vercel:

1. Push the frontend files to GitHub.
2. Import the frontend repository in Vercel.
3. Deploy the project.
4. Make sure `public/assets/js/env.js` points to the correct backend API URL.
5. Redeploy after changing `env.js`.

InvestSmart Installation and Deployment Guide

6. Hard refresh browser with Ctrl + F5 when testing.

14. Troubleshooting:

Failed to fetch:

- Backend URL in env.js may be wrong.
- ngrok may be offline.
- Apache may be stopped.
- CORS/backend may not be returning JSON.

Invalid JSON:

- Backend may be returning an HTML error page.
- PHP error may be displayed instead of JSON.
- API URL may point to the wrong folder.

Cannot login:

- Check that the database exists.
- Check that schema.sql was run.
- Check that auth_otps table exists.
- Check admin email and password.
- Check that OTP is entered after password login.

Buttons not working:

- Hard refresh with Ctrl + F5.
- Confirm public/assets/js/auth.js is loaded.
- Confirm public/assets/js/api.js has the correct API path.

15. Final Submission Checklist:

- Backend code pushed to GitHub.
- Frontend code pushed to GitHub.
- Database schema included.
- ERD included.
- Proposal PDF included.
- Installation/deployment PDF included.
- Test cases PDF included.
- Security information PDF included.
- System tested using client and admin accounts.